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Quick Recap!

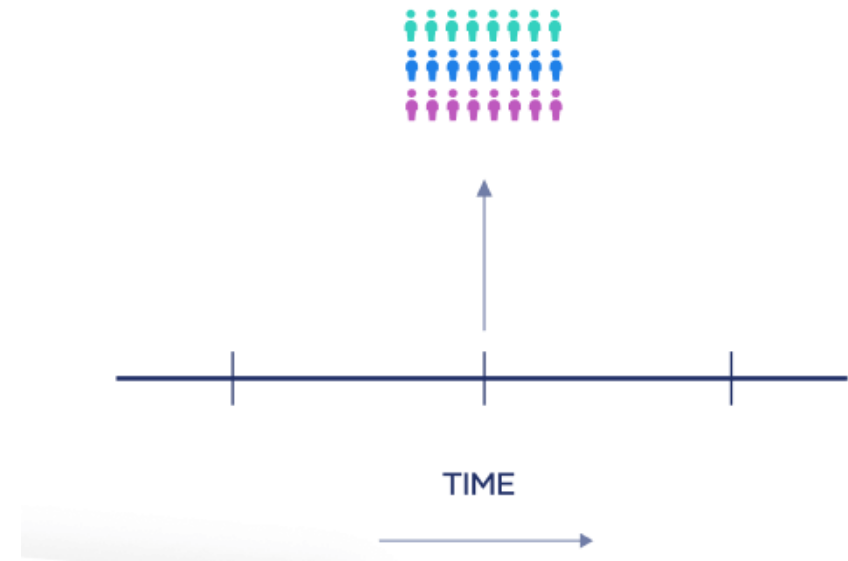
- Scientific Method
- Perceiving the question – Hypothesis – Hypothesis Testing – Conclusion – Report
- Types of research methods
 - Descriptive
 - Correlational
 - Experimental
- Descriptive types
 - Naturalistic observation
 - Laboratory observation
 - Clinical case study
 - Survey

Descriptive Method: Cross-Sectional & Longitudinal Research

- Cross-sectional research: analysing information about a population at a **specific point in time**
- Strength:
 - Cost-effective
 - Can incorporate multiple variables and groups
- Drawbacks:
 - Reporting bias: sharing selective information from the participant and/or researcher
 - Single time point: difficult to understand long term trends or analyse change over time

Cross-sectional study

Data collected at one point in time



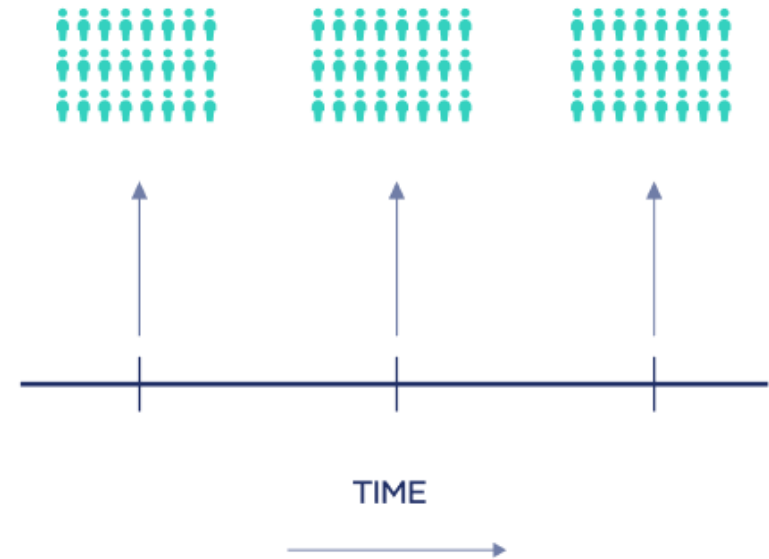
(Simply Psychology, 2023)

Descriptive Method: Cross-Sectional & Longitudinal Research

- Longitudinal research: analysing information about a specific population over an **extended period of time**
- **Retrospective study:** you collect data on events that have already happened
- **Prospective study:** you choose a group of subjects and follow them over time, collecting data in real time
- Strength:
 - Detect patterns and changes in the population
- Drawbacks:
 - Expensive and time-consuming

Longitudinal study

Data collected repeatedly over time



(Simply Psychology, 2023)

Descriptive Method: Cross-Sectional & Longitudinal Research (cont'd...)

Research Type	What It Means	Example
Cross-sectional research	Looks at different groups at one moment in time	Surveying high schoolers, college students, and new grads <u>all in 2025</u> to see who uses Instagram the most
Longitudinal research	Studies the same group of people over a period of time	Tracking the same group of Gen Z teens <u>from 2020 to 2025</u> to see how their favorite social apps change



Cross-sectional research is like taking a group selfie: everyone at once, different ages or groups



Longitudinal research is like a series of throwback selfies: same people, but over a few years

Correlational Research: Finding Relationships

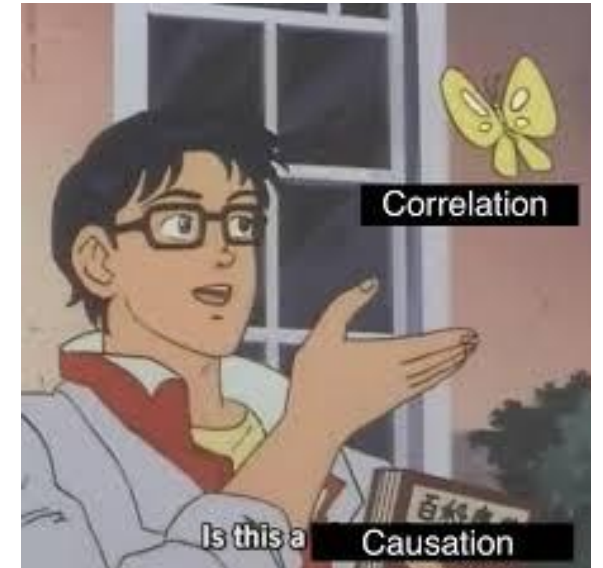
- Correlational research is a statistical technique
- Looks for **patterns** in the data
- A measure of the **relationship** between two or more variables
- **Variable**: anything that can change or vary



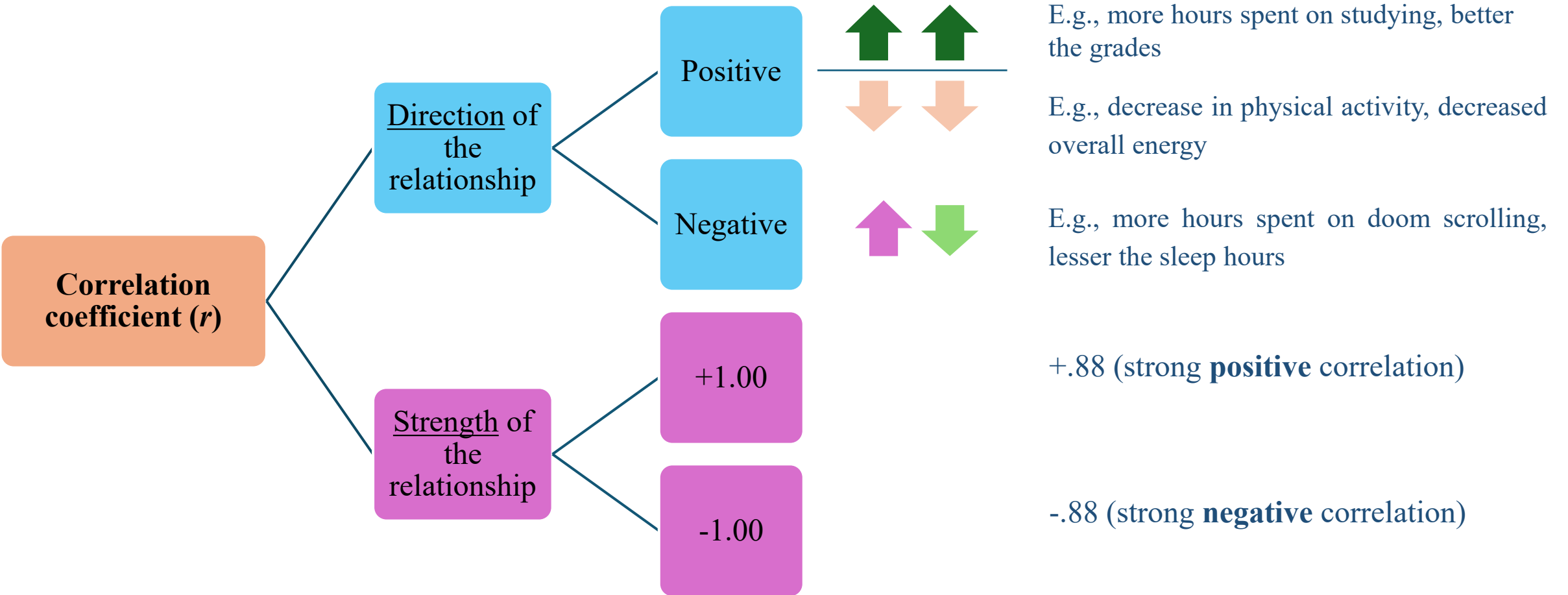
E.g.,

- Ice cream sales and drowning both increase in the summer
- Students who spend more time on social media often feel more anxious

Correlation does not imply causation!



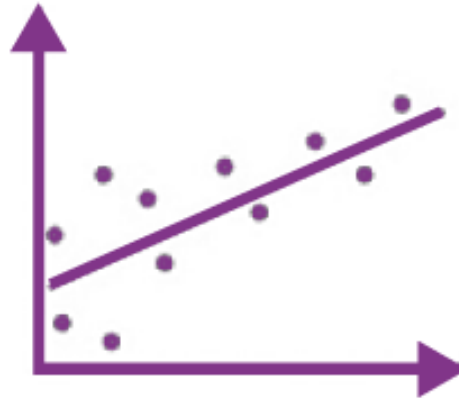
Finding Relationships: Correlational Research (cont'd...)



✨ Closer to zero: weaker relationship ✨



Strong positive correlation



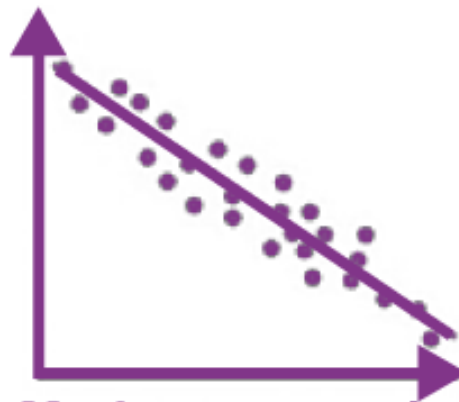
Weak positive correlation



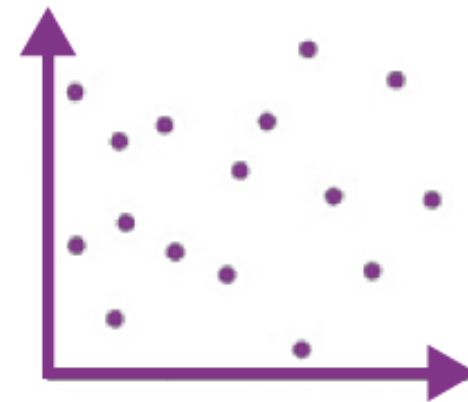
Strong negative correlation



Weak negative correlation



Moderate negative correlation



No correlation

Quick Recap!

Cross sectional research

- Observational study that collects data from diverse population at a single point in time
- Helps understand prevalence and describe characteristics

Longitudinal research

- A study method that involves collecting data from same individual or group over extended period of time from weeks to decades

Correlational research

- Non experimental method that measures the statistical relationship between two or more variables
- It seen if change in one corresponds change in another without manipulation